Physics 3221 Spring 2026: Practice Midterm 2

Grade:

1. Mystery Force and its Potential (30pts)

Consider the following force: $\vec{F} = ky\hat{x} + kx\hat{y}$.

(a) Calculate $\vec{\nabla} \times \vec{F}$. Is this force conservative?

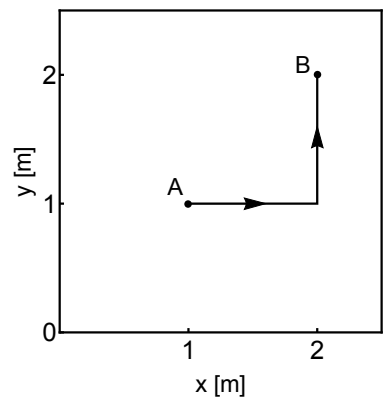


Figure 1: The path the particle follows going from point A to point B

(b) Find the W the force does on a particle moving from point A to point B along the path depicted in Fig. 1.

(c) For this force, $U(x, y) = -xy$. Verify that $\vec{F} = -\vec{\nabla}U$

Grade:

2. The Underwater Clock

A scuba diver finds a submerged clock sitting inside a sunken ship. The clock tells time using a swinging pendulum, see Fig. 2. Assume the mass attached to the pendulum experiences a gravitational force with force constant g, and ignore the buoyancy force.

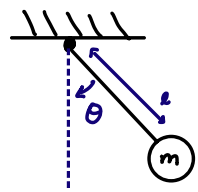


Figure 2: The pendulum inside the underwater clock. The length of the pendulum arm is l and the mass attached is m.

(a) Neglecting drag, show that the potential energy of the mass on the pendulum for small angles $\theta \ll 1$ is $U(\theta) \approx \frac{1}{2}k\theta^2$. Find k in terms of m and l.

(b) Find the net force acting on the mass in the angular direction, again assuming small angels and ignoring terms of $\mathcal{O}(\theta^3)$.

(c) Write down the equation of motion for the pendulum, assuming $\theta \ll 1$.

(d) Find $\theta(t)$. Use the following initial conditions: at $t = 0$, the pendulum starts at an angle θ_0 and is released from rest.

(e) What is the frequency and amplitude of oscillations?

Grade:

3. The Damped Underwater Clock

Now, consider the effect of drag from the water on the mass in the pendulum shown in Fig. 2. The water exerts a drag force $\vec{F}_{\text{drag}} = -b\vec{v}$.

(a) Show that the new equation of motion for the pendulum is:

$$\ddot{\theta} + 2\beta\dot{\theta} + \omega_0^2\theta = 0$$

and find ω_0 and β.

(b) Solve the equation of motion to find $\theta(t)$, assuming $\beta \neq \omega$. Use the same initial conditions as the previous problem: at $t = 0$, the pendulum is released from rest at a position θ_0 away from the vertical axis.

(c) What are the conditions on b to ensure that this pendulum oscillates? Write your answer in terms of m, g, and l.

(d) What are the conditions on b to ensure that would mean this pendulum is not oscillating? Write your answer in terms of m, g, and l.

(e) If $\beta = \omega_0$, find the solution for $\theta(t)$ using the same initial conditions.

Grade:

4. The Driven Damped Underwater Clock

Again consider that the water exerts a drag force on the pendulum shown in Fig. 2. This time, the pendulum is also subject to a driving force $F(t)$ in the $\hat{\theta}$ direction.

(a) Write down the equation of motion for the clock given both the drag and driving forces.

(b) Let $F(t) = F_0 \cos(\omega t)$ with the same initial conditions as the previous problem: at $t = 0$, the pendulum is released from rest at a position θ_0 away from the vertical axis.

i. Find $\theta(t)$.

ii. Find the value of ω that gives a resonance for this driven oscillator, assuming ω_0 is fixed.

(c) Now consider instead that the driving force is periodic with the following shape:

$$F(t) = F_0 t, \qquad t \in (-\tau/2, \tau/2) \tag{1}$$

$$F(t+\tau) = F(t) \tag{2}$$

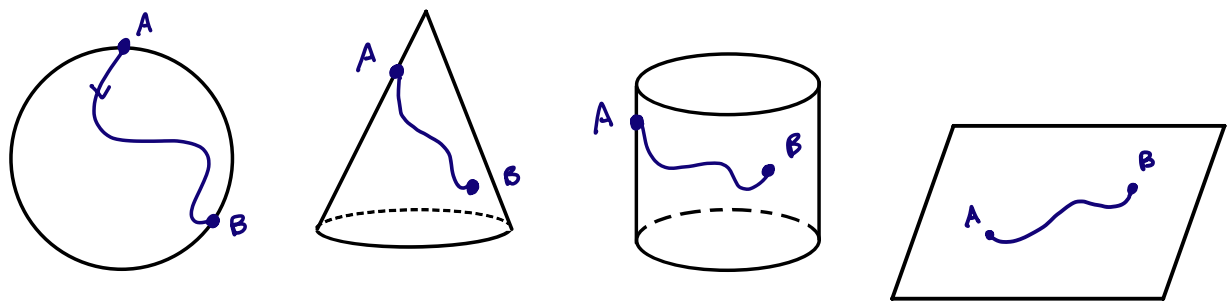
i. Sketch the shape of $F(t)$

ii. Find the Fourier expansion of $F(t)$

iii. Find the general solution for $\theta(t)$ given the driving force shown in Eq. (2).

Grade:

5. **Geodesics** Calculate the geodesic (the shortest distance between two points) of a few surfaces: cone, sphere, cylinder, plane



(a) A sphere with radius R

(b) A circular cone with a surface defined by $\rho = \lambda z$

(c) A cylinder of radius R

(d) A flat plane

Figure 3: Several surfaces

(a) Find the shortest distance between point A at position $(R, 0, 0)$ and B at position (R, θ_B, ϕ_B) following a path on the surface of a sphere shown in Fig. 3a.

(b) Find the shortest distance between point A at position (ρ_A, π, z_A) and B at position (ρ_B, ϕ_B, z_B) following a path on the surface of a cone shown in Fig. 3b. Once you get to trying to determine the two unknown coefficients by plugging in ϕ_b, z_b, z_A, you can stop, as the expressions get complicated.

(c) Find the shortest distance between point A at position (R, π, z_A) and B at position (R, ϕ_B, z_B) following a path on the surface of the cylinder shown in Fig. 3c.

(d) Find the shortest distance between point A at position $(0, 0, 0)$ and B at position (x_B, y_B, z_B) following a path on the surface of a cone shown in Fig. 3d.

Grade:

6. Things to Practice:

- Practice Line Integrals
- Practice Finding $U(\vec{r})$ from $\vec{F}(\vec{r})$ and vice versa
- Practice Checking Whether $\vec{F}$ is Conservative using Curl
- Practice Using Spherical Coordinates
- Practice Solving SHO 2nd order differential equations
- Practice Finding Fourier Series for various functions
- Practice using calculus of vtariations to minimize actions

Formulas (just for the Practice Midterm)

Newton's 2nd Law

$$\vec{F} = m\ddot{\vec{r}}$$

Cartesian

$$F_x = m\ddot{x} \qquad F_y = m\ddot{y} \qquad F_z = m\ddot{z}$$

Polar

$$F_r = m(\ddot{r} - r\dot{\phi}^2) \qquad F_\phi = m(r\ddot{\phi} + 2\dot{r}\dot{\phi})$$

Cylindrical

$$F_\rho = m(\ddot{\rho} - \rho\dot{\phi}^2) \quad F_\phi = m(\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi}) \quad F_z = m\ddot{z}$$

Coordinate Systems

Cartesian: $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$d\vec{r} = dx\hat{x} + dy\hat{y} + dz\hat{z}$

Cylindrical : $\vec{r} = \rho\hat{\rho} + z\hat{z}$

$d\vec{r} = d\rho\hat{\rho} + \rho d\phi\hat{\phi} + dz\hat{z}$

Spherical: $\vec{r} = r\hat{r}$

$d\vec{r} = dr\hat{r} + rd\theta\,\hat{\theta} + r^2\sin\theta\, d\phi\,\hat{\phi}$

Euler's Formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Fourier Series

$$f(t) = \sum_{n=0}^{\infty}\left[a_n\cos(n\omega t) + b_n\sin(n\omega t)\right]\,,$$

$$\omega = 2\pi/\tau.$$

$$a_0 = \frac{1}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\,,$$

$$a_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\cos(n\omega t)dt\,, \qquad n > 0$$

$$b_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\sin(n\omega t)dt\,, \qquad n > 0$$

Hook's Law & Simple Harmonic Motion

$$F = -kx \Leftrightarrow U = \frac{1}{2}kx^2$$

$$x(t) = A\cos(\omega t - \delta)$$

Driven Oscillator

$$A^2 = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2)}$$

$$\delta = \tan^{-1}\left(\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right)$$

Lorentz Force Law

$$\vec{F} = q(\vec{E} + \vec{v}\times\vec{B})$$

Work

$$W = \int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}\,')\cdot d\vec{r}\,'$$

Conservative forces

$$U(\vec{r}) = -\int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}\,')\cdot d\vec{r}\,'$$

$$F = -\vec{\nabla}U$$

$$\vec{\nabla}\times\vec{F} = 0$$

Useful Integrals

$$\int x\sin(ax)dx = -\frac{x}{a}\cos(ax) + \frac{1}{a^2}\sin(ax)$$

$$\int x\cos(ax)dx = \frac{x}{a}\sin(ax) + \frac{1}{a^2}\cos(ax)$$

Gradient and Curl

$$\vec{\nabla}f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$

$$\vec{\nabla}f = \hat{r}\frac{\partial f}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial f}{\partial\theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial f}{\partial\phi}$$

$$\vec{\nabla}f = \hat{\rho}\frac{\partial f}{\partial r\rho} + \hat{\theta}\frac{1}{\rho}\frac{\partial f}{\partial\theta} + \hat{z}\frac{\partial f}{\partial z}$$

$$\vec{\nabla}\times\vec{A} = \hat{x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$